PIPERACILLIN + TAZOBACTAM

Antipseudomonal
Penicillin allergy

CHEMICAL NAME Piperacillin (ureidopenicillin) + tazobactam: (2S,3S,5R)-3-methyl-7-oxo-3-(1H-1,2,3-triazol-1-ylmethyl)-4-thia-1-azabicyclo[3.2.0]heptane-2-carboxylic acid 4,4-dioxide

CLASS Extended-spectrum (antipseudomonal) penicillin + β-lactamase inhibitor

TRADE NAMES Tazocin · Zosyn · Piptaz

THERAPEUTIC USES / INDICATIONS

- Hospital-acquired and ventilator-associated pneumonia.
- Complicated intra-abdominal infections, peritonitis.
- Febrile neutropenia (empirical).
- Complicated UTI, skin and soft-tissue infections, diabetic foot; sepsis of unknown source.

ADMINISTRATION

- **IV infusion** over 30 min (or extended infusion over 4 h to improve efficacy).
- Not compatible with aminoglycosides in same line — give separately.
- Each 4.5 g vial contains sodium (~9.4 mmol) — note in heart failure.

DOSAGES

- **Adult:** 4.5 g IV every 8 h (every 6 h in severe infection/neutropenia).
- **Children (>2 months):** 90 mg/kg every 6–8 h (max 4.5 g/dose).
- **CrCl 20–40 mL/min:** 4.5 g every 8 h; **<20:** 4.5 g every 12 h.
- Review at 48–72 h with culture results.

PHARMACOKINETICS

- **Onset:** immediate (IV). Good tissue penetration.
- Minimal metabolism.
- Excreted in urine. **t½:** ~1 h.

PHARMACODYNAMICS

- Piperacillin inhibits **cell-wall synthesis** (bactericidal), including *Pseudomonas aeruginosa*.
- Tazobactam inhibits many **β-lactamases** → covers Gram-negatives, anaerobes, MSSA.
- Not active against MRSA or ESBL-resistant in many cases.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Diarrhoea, nausea, vomiting.
- Rash, itching.
- Headache, insomnia; phlebitis.

ADVERSE EFFECTS (SERIOUS)

- Anaphylaxis.
- *C. difficile* colitis.
- Thrombocytopenia, bleeding (prolonged courses); leucopenia.
- Acute kidney injury (especially combined with vancomycin); hypokalaemia; seizures (high dose).

CONTRAINDICATIONS & CAUTIONS

- Hypersensitivity to penicillins, cephalosporins or β-lactamase inhibitors.
- **Caution:** renal impairment, sodium restriction, bleeding disorders.

KEY INTERACTIONS

- **Vancomycin:** ↑ nephrotoxicity.
- **Aminoglycosides:** inactivation if mixed.
- **Methotrexate:** ↑ toxicity.
- **Vecuronium:** prolonged block.

NURSING MANAGEMENT

- Check allergy history before first dose.
- Take cultures before starting; give first dose promptly in sepsis.
- Monitor renal function, K^+, CBC, coagulation in long courses.
- Monitor temperature and clinical response; prompt review at 48–72 h.
- Watch for diarrhoea.

PATIENT EDUCATION

- Report rash, itching, bleeding or bruising.
- Report watery or bloody diarrhoea, even after the course ends.

KEY POINTS

- Broad spectrum incl. Pseudomonas and anaerobes.
- Febrile neutropenia and hospital pneumonia.
- 4.5 g IV every 8 h (6 h if severe).
- AKI risk with vancomycin.